DAHD example: identification of spatio-temporal oscillatory modes in noisy data

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1. Numerical setup

To demonstrate DAHD capabilities, we use synthetic dataset at http://research.atmos.ucla.edu/tcd/ssa/guide/mssa/mssarot.html, as part of the SSA-MTM Toolkit for time series analysis. We utilize more accurate and computationally-efficient DAHD formulation in frequency domain (Kondrashov et al., 2020, 2026) rather than its original time domain formulation (Chekroun and Kondrashov, 2017). Here DAHD is realized numerically by an eigendecomposition of the Hermitian cross-spectral density matrix (Kondrashov et al., 2020, 2026). The resulting eigenvectors yield in physical space an orthonormal set of spatiotemporal data adaptive harmonic modes (DAHMs), associated with a specific temporal frequency, which in turn allows to rank these modes by their captured energy and across frequencies, and allow accurate space-time reconstruction.

We consider a short and noisy spatio-temporal dataset describing the time evolution of a d-dimensional vector $\mathbf{y}(t_n) := (y_1(t_n), ..., y_d(t_n))$ over the interval $n = 1, ..., N$; here $d = 6$ and $N = 129$. The full dataset shown in Fig.1f consists of a coherent component $\mathbf{s}(t)$ embedded into *temporally correlated*, albeit spatially uncorrelated noise $\mathbf{r}(t)$:

$$\mathbf{y}(t_n) = (1-\nu)^{1/2}\,\mathbf{s}(t_n) + \nu^{1/2}\,\mathbf{r}(t_n)\,. \tag{1}$$

The coherent component $\mathbf{s}(t)$ in Fig.1e is the sum of the four oscillatory modes $x_k^i(t)$ with varying amplitude and phase across the six spatial channels, as shown in Figs.1(a–d):

$$s_k(t_n) = \sum_{i=1}^{4} x_k^i(t_n), \quad k = 1, \cdots, 6\,; \tag{2}$$

these modes are given by

$$x_k^i(t) = \left(\frac{\alpha_k^i}{2}\right)^{1/2} \sin(2\pi f_i t + \Phi_k^i), \quad k = 1, \cdots, 6, \quad i = 1, \cdots, 4, \tag{3}$$

and each phase Φ_k^i is obtained independently as a random variable uniformly distributed in $[0, 2\pi]$.

The periodicities of the four oscillatory modes are not integer multiples of the sampling time nor of each other, while the respective frequencies $f_1 = 1/7.5$, $f_2 = 1/5$, $f_3 = 1/2.8$ and $f_4 = 1/2.3$ (in sampling units) are located in both the low-frequency and high-frequency part of the power spectrum. The amplitudes α_i^j are prescribed across the spatial channels so that 3 distinct modes contribute to each channel, albeit with different amplitudes; see Table 1. The random choice of the

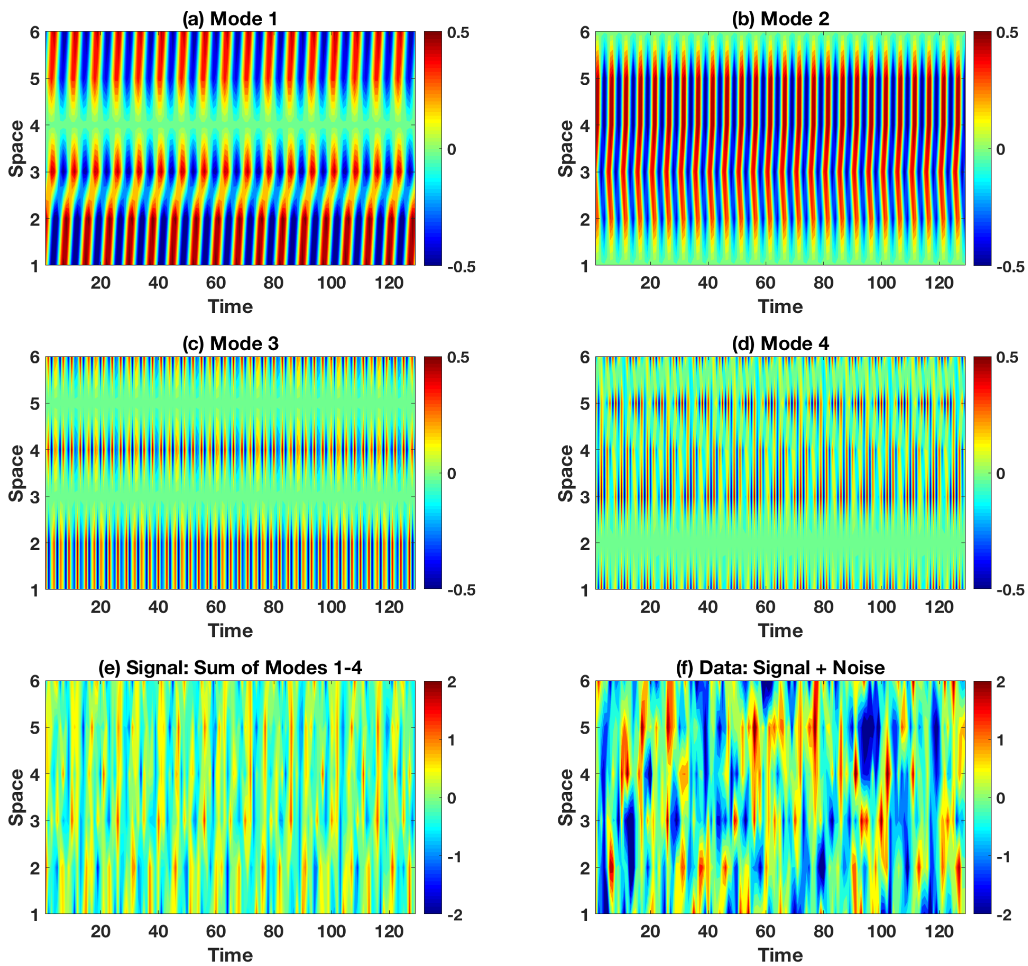

Figure 1: Multivariate spatio-temporal dataset representing six channels in space and 129 points in time: (a–d) four harmonic modes $\{\mathbf{x}^i(t) : i = 1, \ldots, 4\}$ having fixed temporal frequencies but different amplitudes and phases in each of the six channels; see Eq. (3). Their sum $\mathbf{s}(t)$ defines the coherent component given by Eq. (2) shown in panel (e); (f) total dataset representing the sum of the coherent component $\mathbf{s}(t)$ and of the temporal red noise $r_k(t)$ in each of the $\{k = 1, \ldots, d\}$ channels; see text for details.

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Table 1: Amplitude modulation of the four oscillatory modes across six spatial channels; see Eq. (3). The index k is for the channels, while the index i is for the modes.

α_k^i	$i=1$	$i=2$	$i=3$	$i=4$
$k=1$	0.4	0.0	0.3	0.3
$k=2$	0.4	0.2	0.4	0.0
$k=3$	0.3	0.3	0.0	0.4
$k=4$	0.0	0.4	0.4	0.2
$k=5$	0.2	0.4	0.0	0.4
$k=6$	0.3	0.0	0.4	0.3

phases Φ_k^i in Eq. (3) results in arbitrary phase shifts across the spatial channels; see Fig. 1. The coefficient $\nu = 0.6$ in Eq. (1) guarantees that the noise component has larger variance than the signal; this fact is obvious from a comparison of the "clean" Fig. 1e with the "noisy" Fig. 1f), and it makes the identification problem that much more challenging.

The embedding window M is the main parameter of DAHD. The choice of M is based on two competing goals: (i) to obtain reliable estimates of autocorrelations from noisy and short datasets; and (ii) to resolve the dataset's frequency domain for identification purposes with sufficient accuracy. M should be larger than typical decorrelation times in the data; it is the slowest temporal scale characterized by DAHD. We chose here maximum spectral resolution $M = (N+1)/2$ or $M = 65$, which results in a total number $dM = 390$ of DAHD eigenvalues λ_j.

Each of the DAHMs $\mathbf{E}_j$, i.e. $1 \leq j \leq dM'$, where $M' = 2M-1$, represents a data-adaptive spatio-temporal pattern associated with a fixed temporal frequency; the latter are equally spaced at intervals of $1/(M'-1)$ in the Nyquist interval $[0, 0.5]$. Moreover, each temporal frequency is associated with d DAHM pairs and eigenvalues, except at $f = 0$, where there is only one DAHM per eigenvalue. Figure 2 shows the DAHD spectrum composed of the values λ_j (red full circles), and obtained here for the synthetic dataset in Fig. 1f. The frequencies of the oscillatory modes that make up the coherent component are identified by DAHD eigenvalues located above the noisy background, and marked by the distinct colors.

The time-embedded structure of the DAHMs associated with a top spectral eigenvalue at these frequencies, are shown in Fig. 4. This structure conveys information about the amplitude modulation across spatial channels, and the figure demonstrates that indeed the DAHMs for each pair, except at zero frequency, are in phase quadrature, i.e. shifted by one quarter of the associated period.

The latter property is reminiscent of Fourier decomposition, based on sine and cosine pairs with the same periodicity, as well as of the similar property of oscillatory SSA eigenpairs (Ghil et al., 2002). The k-th spatial channel $\mathbf{E}_k^j$ of a particular multivariate DAHM — i.e., for a DAHD with $d \geq 2$ — that is associated with a frequency

$$\omega_\ell = \frac{2\pi(\ell-1)}{M-1}, \quad \ell = 1, \cdots, \frac{M+1}{2}. \tag{4}$$

can be analytically expressed—for each $1 \leq j \leq dM'$—as an oscillatory function in the embedding time-window variable τ as follows:

$$\mathbf{E}_k^j(\tau) = B_k^j(\omega_\ell)\sin(\omega_\ell \tau + \phi_k^j(\omega_\ell)), \quad 1 \leq k \leq d,\ 1 \leq \tau \leq M'\,; \tag{5}$$

here both amplitudes $B_k^j(\omega_\ell)$ and phases $\phi_k^j(\omega_\ell)$ are *data-adaptive*.

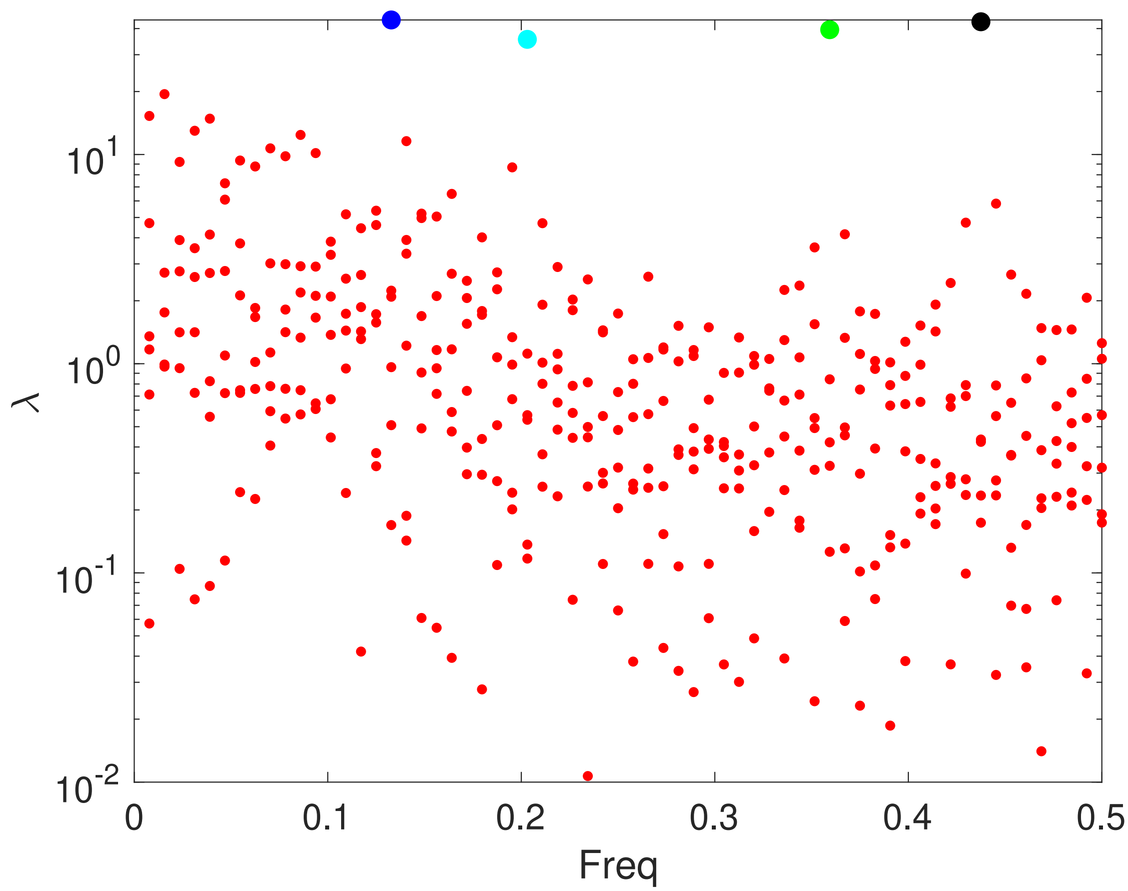

Figure 2: DAHD spectrum of the noisy dataset in Fig. 1f. Each red full circle corresponds to a λ_j with pair of data-adaptive DAHD modes (DAHMs) $(\mathbf{E}_j, \mathbf{E}'_j)$; the latter represent the same temporal frequency f_j but are time-shifted so as to be in phase quadrature, cf. Fig. 4 below. The frequencies of the DAHMs are equally spaced between 0 and 0.5, and the total number of DAHM pairs in each frequency bin is equal to the number of channels $d = 6$ in the dataset. The DAMHs describe amplitude and phase modulation between the spatial channels and are shown in Fig. 4. Faithful reconstruction of the reference modes of coherent component in Figs. 1(a–d), as shown in Figs. 5(a–d) below, is obtained by using top spectral elements above the noisy background, an marked by distinct colors: blue – f_1, cyan – f_2, green – f_3 , and black – f_3.

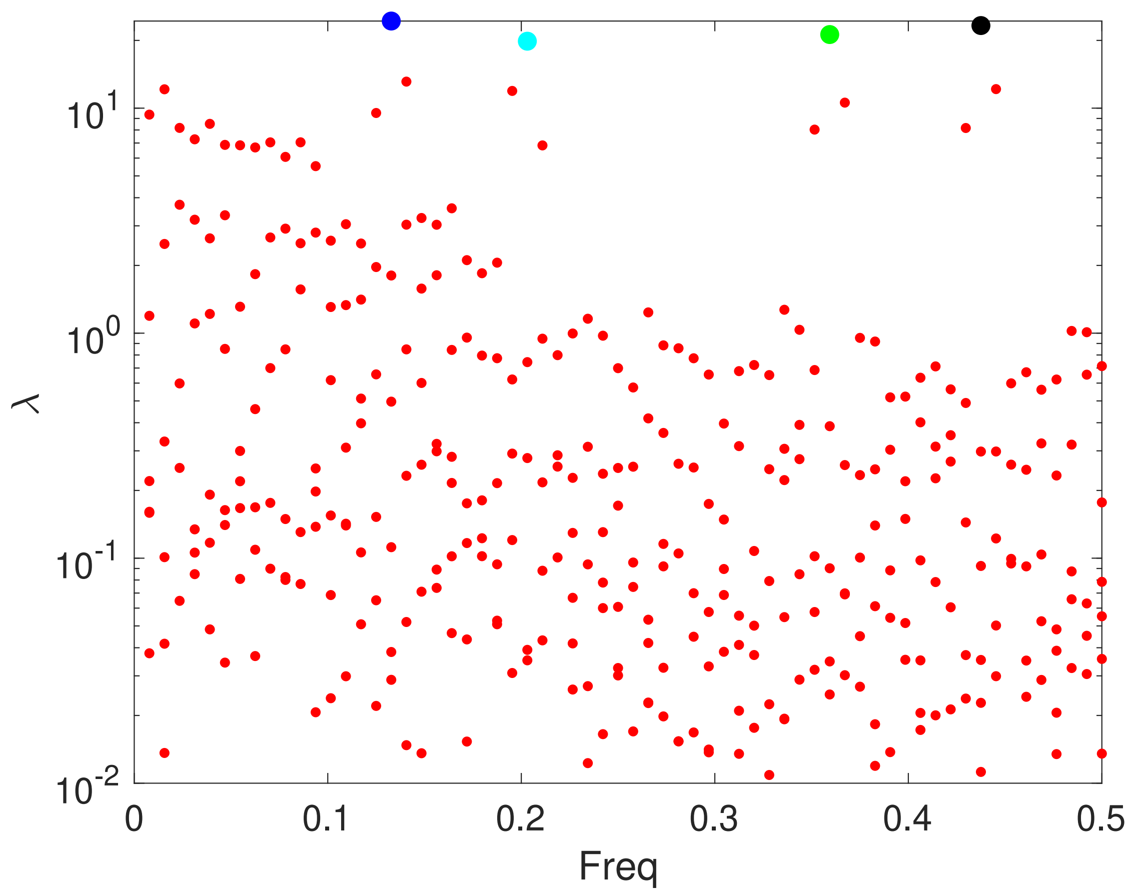

Figure 3: Same as in Fig. 2 but with weighting applied to cross-correlations.

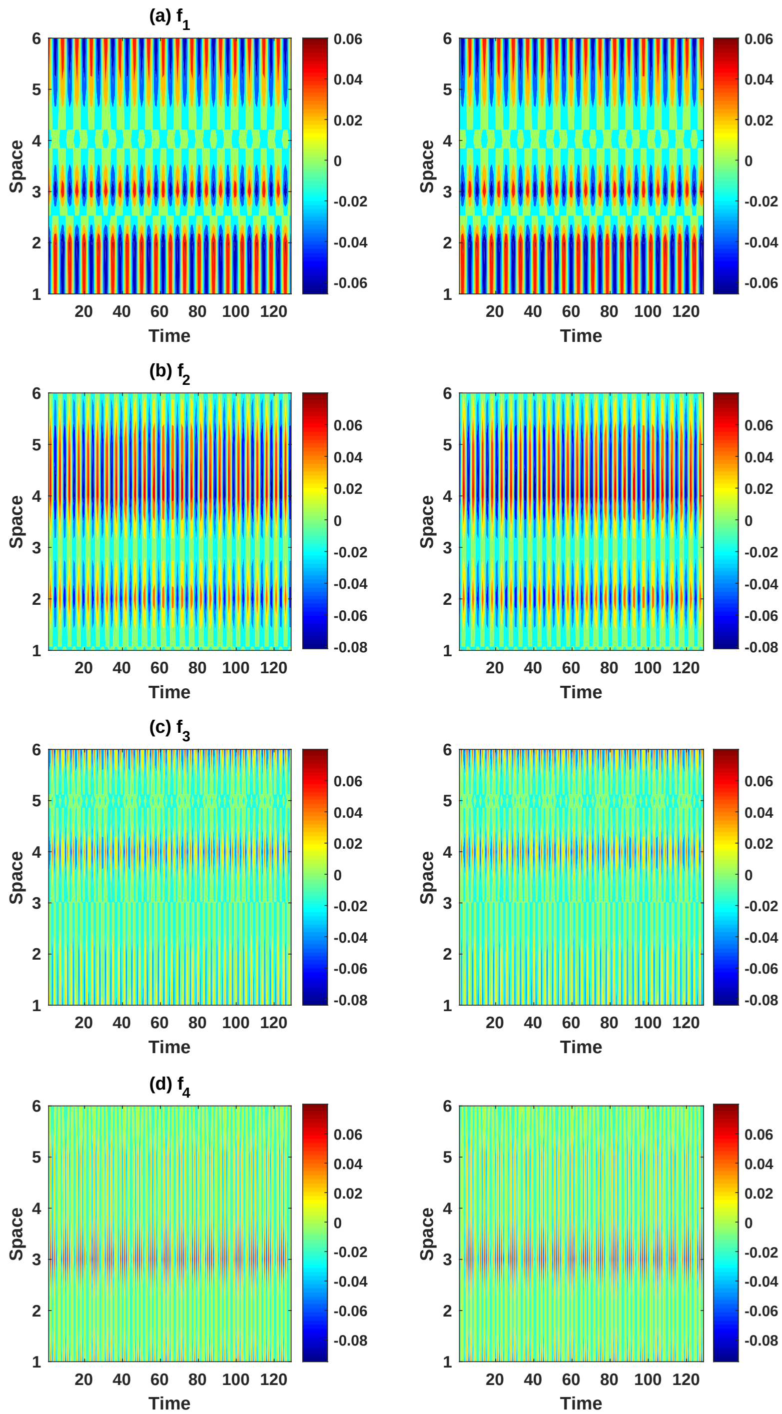

Figure 4: $(\mathbf{E}_j, \mathbf{E}'_j)$ of the leading spectral DAHM pairs for the four frequencies of the four spatio-temporal oscillatory modes in Figs. 1(a–d), i.e. f_1, f_2, f_3 and f_4, respectively; see Eq. (3). The x-axis represents the embedding dimension M', while the y-axis corresponds to $d = 6$ spatial channels. For each spatial channel, the modes of a given frequency convey different phases and are shifted by a quarter of the associated period, i.e. they are in exact phase quadrature.

Moreover, the modes in each pair are shifted by one fourth of the period, i.e. DAHMs are in exact phase quadrature, as for sine–cosine pairs, but in a data-adaptive fashion. The precise information about amplitude and phase modulation of the oscillatory modes captured by the DAHMs allows one to perform highly accurate reconstructions in the space-time domain (Chekroun and Kondrashov, 2017; Kondrashov et al., 2018, 2020). Figure 5 shows the space-time patterns of the harmonic reconstruction components (HRCs); these patterns are obtained by using two leading DAHD pairs in the frequency bins that contain the target periodicities f_1, f_2, f_3 and f_4, see Fig. 2. As we can see these patterns match quite well in frequency and phase those of the reference coherent components. These results show that DAHD can correctly identify distinct oscillatory modes in a very noisy multichannel dataset, providing considerable improvement over capabilities of varimax-rotated M-SSA algorithm (Groth and Ghil, 2011), see `http://research.atmos.ucla.edu/tcd/ssa/guide/mssa/mssarot.html`.

2. Matlab routines

dahdexample4.m – main script of this example.
generate_data.m– auxiliary routine to generate input dataset of this example.
DAHD4freq_part_weight.m – DAHD power spectrum and eigenvectors
DAHM4_ex.m –computes space-time DAHMs from DAHD eigenvectors.
dahc.m –computes DAHD coefficients (DAHCs) from DAHMs and input dataset.
hrc.m –computes HRC by using DAHCs and DAHMs.

3. Best practices/FAQ

- Embedding window M' should be larger than typical decorrelation times in the data; it is the slowest temporal scale characterized by DAHD. Note that for this example M' is chosen to its maximum value, which is half of the length N of the very short dataset, and such case DAHCs become just scalar values. For the very long time series it is practical and useful to use $M' << N$. The resulting DAHCs of length $N - M' + 1$ can be then simulated by the model of coupled stochastic Multi-level Stuart-Landau oscillators (MSLM) (Kondrashov et al., 2018).

- The Hermitian DAHD formulation (Kondrashov et al., 2020, 2026) closely resembles a SPOD/SEOF approach (Towne et al., 2018), with the key difference being that the latter considers Welch's overlapped averaging periodogram estimate of the cross-spectral density (CSD) matrix, while DAHD is based on a correlogram estimate; both in turn can be understood as approximating dynamic mode decomposition (DMD) in ensemble context and relating to Koopman theory (Tu et al., 2014).

- Applying weighting to time-lagged cross-correlations prior to computing CSD matrix reduces bias at large lags, helps to make it positive-define and its eigenvalues strictly positive.

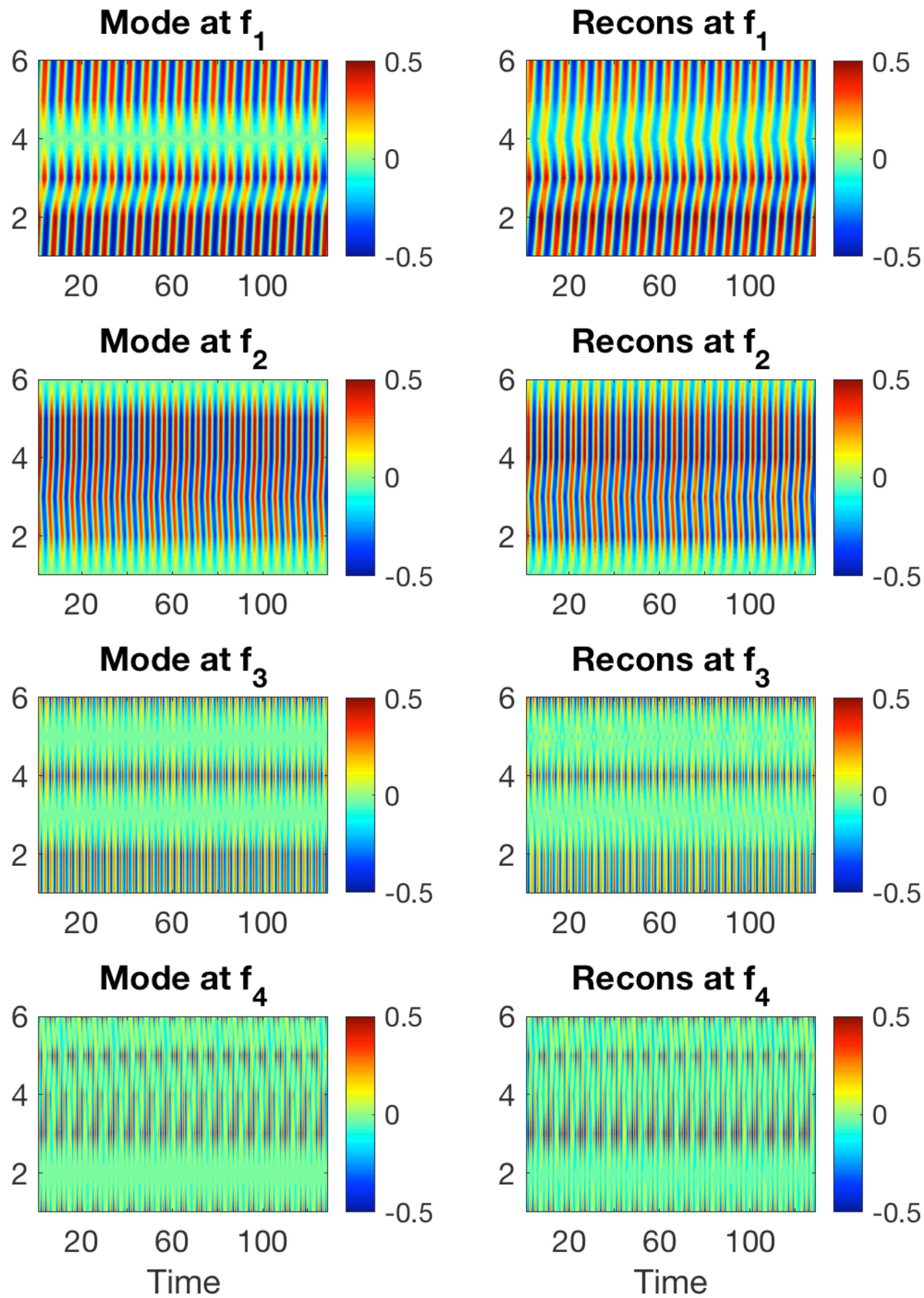

Figure 5: DAHD reconstruction (right panel) obtained by using two leading DAHD pairs in the corresponding frequency bins (colored dots in Fig .2). The resulting patterns match reasonably well the reference patterns (left panel).

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